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PAPER TITLE:**

“Is Foreign Aid to the Energy Sector Effective in Increasing Electricity Access?”

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1. Introduction

A key debate on the topic of foreign aid is on the effectiveness of aid with voluminous literature in this field. From 2010 to 2020, on average 70 to 90 papers were released a year on aid effectiveness, which is up from around 30 papers on average per year between 2000 and 2010 (Asatullaeva et al. 2021). A key driver in this field of research is the Sustainable Development Goals (SDGs). The SDGs are 17 goals agreed by UN members in 2015 that act as the blueprint for a more sustainable future targeting 2030 (United Nations 2023). The goal is to leave no one behind, hence “development aid will play a central role in the global strategy for reaching this goal” (Haga and Høybråten 2010). As a result, aid effectiveness in reaching these SDGs became a major avenue of research. See Jakubik and Feuerriegel 2022; Faust 2018; and Seelkopf and Bastiaens 2020 for some recent reviews about aid effectiveness on SDGs.

In this paper, we examine aid effectiveness on the 7th SDGs, which is “about ensuring access... to affordable energy” (United Nations 2023). We draw data from the Organization for Economic Co-operation and Development’s Development Assistance Committee (OECD-DAC) and the World Bank to run our country fixed effects model estimating the impact of aid on overall, rural, and urban access to electricity for countries who received ODA to their energy sector between 2013 and 2021. The main takeaway from the results is that aid has a positive effect on access to electricity at both contemporary and lagged channels.

2. Related Literature

Seminal papers on the topic suggest mixed evidence regarding the efficacy of foreign aid. Qian (2015) surveys the evidence, demonstrating the lack of consensus regarding aid effectiveness, with unintended consequences such as the potential for moral hazard hindering positive impacts. Burnside and Dollar (2000) argue that conditional on ‘good’ policies, such as ones which facilitate budget surpluses, low inflation and trade openness, aid can have significant positive effects on growth. However, Easterly (2003) rebukes these results arguing that they are not robust to greater data inclusion and alternative time periods. Instead, Easterly contends that aid can only be beneficial in specific cases as opposed to the ‘aggregate’.

More recently, academic literature has begun to concentrate on evaluating the efficacy of aid disbursement to specific sectors. Williamson (2008) studied the impact of foreign aid disbursements to the health sector using a series of fixed-effect models, finding that foreign aid caused no significant improvement to the aggregate health of the recipient nations. However, when analysing the impacts of aid disbursements to education, Yogo (2017) ran a 2SLS regression and found that aid disbursements to primary education significantly increased the primary education completion rate, robust to various specifications. Gopalan and Rajan (2016) use panel data and a country fixed-effects model to examine the effects of foreign aid

disbursements on the water and sanitation supply (WSS) sectors, finding a positive and significant effect of aid on WSS indicators, robust to multiple specifications.

Our paper contributes to the discussion of foreign aid by leveraging the methods in Gopalan and Rajan (2016) to examine the effectiveness of aid disbursements to the energy sector. This sector is crucial to a nation's development as energy poverty (the reliance on primitive sources of energy) is linked to negative individual health and learning outcomes (Xiao et al 2023), and lower economic development (Amin et al 2020). This highlights the significance of this avenue of research.

3. Data and Methodology

Using the OECD-DAC and the World Bank Development Indicators database, we compiled panel data for 114 countries receiving ODA from the DAC from 2013 – 2021. Our chosen dependent variable measures the percentage of a country that has access to electricity, whilst our independent variable captures the volume of ODA disbursement to the energy sector received by country and year as shown by the OECD-DAC database. Countries were eligible for inclusion if they received ODA to their energy sector from the OECD between 2013 and 2021, and they had sufficient data.

3.1 empirical model

Our model seeks to estimate the change in access to electricity within a country associated with changes to the volume of ODA disbursements to the energy sector of the recipient developing country. Subsequently, we look to draw inferences regarding the effectiveness of ODA to the energy sector in improving electricity access. We run a country fixed effects model, and our base equation is as follows:

$$AE_{it} = \alpha_i + \beta Aid_{it} + \delta \mathbf{X}_{it} + \varepsilon_{it} \quad (1)$$

Here, AE_{it} refers to the percentage of the population that has access to electricity in country i at time t . This is our dependent variable of interest. We considered several others, most notably '*Time required to get electricity*', but chose '*Access to electricity*' as it had the most variation and data availability across countries and time. We worked with '*Access to electricity*' for the overall population initially, but subsequently also ran separate regressions for urban and rural populations to investigate heterogeneity.

Aid_{it} represents the gross aid disbursement to the energy sector in country i at time t . We run three variations of this term to capture the channels in which aid may affect electricity access. These include: $LnAid$ – gross ODA disbursements to the energy sector expressed as natural logarithms, $LnAid/Population$ – ODA per Capita, and $LnAid_{t-1}$ – where we lagged $LnAid$ by one year to account for time lags and address joint endogeneity concerns.

Other terms include α_{it} – the country fixed effect, ε_{it} – the idiosyncratic error term, and $\mathbf{X}_{it}$ – a vector of control variables. These control variables can be split into three groups: macroeconomic, demographic, and institutional indicators. All data for

control variables come from the World Bank Development Indicators database¹. The specific variables in each group are as follows:

1. **Macroeconomic** – ‘*GNI per Capita*’, ‘*Inflation*’ (expressed as natural logarithms), ‘*Total Government Expenditure as a percentage of GDP*’, ‘*Net Inflows of Foreign Direct Investment, as a percentage of GDP*’, and the ‘*KOF Globalisation Index*’ as a proxy for trade openness.
2. **Demographic** – ‘*Population Density* (people per km² of land area)’, and ‘*Age Dependency Ratio* (number of dependents per 100 people of working age)’
3. **Institutional** – ‘*Government Effectiveness*’, ‘*Regulatory Quality*’, and ‘*Rule of Law*’. These are all indices, ranging from -2.5 (not effective) to 2.5 (very effective).

In addition to running all our regressions with country fixed effects, we cluster robust standard errors to address heteroskedasticity and autocorrelation concerns.

3.2 limitations

The empirical analysis of this paper faces several constraints, which can be organised into two categories: data limitations; and endogeneity concerns.

With regards to data limitations, we encountered difficulty sourcing precise data in developing nations, as well as only having limited access to data across a large time period. This may give rise to measurement errors, a biased sample and results which fail to characterise the relationship between ODA disbursements to the energy sector, and energy outcomes, in a general way. Firstly, though the data was sourced from credible institutions such as the World Bank, OECD, and KOF, there still may be measurement error given the difficulty in obtaining data in developing countries, which may be unstable and conflict-ridden. Secondly, we could not obtain data for the period before 2013, which reduces the scope of our analysis exclusively to this period. This is particularly important as summary statistics for our dependent variable (‘percentage of the population with access to electricity’) reveal that over half of our sample has coverage exceeding 91%. This restricts our ability to study long-term trends and generalise this relationship.

Table 1: Summary Statistics: Sample Distributions of Electricity Access for Overall, Rural and Urban Populations

	Observations	Mean	P1	P5	P10	P25	P50	P75	P90	Max
Electricity Access (% of total population)	1130	75.94742	9.245298	17.5	29.495	52.46192	91.79708	99.71	100	100
Electricity Access (% of Rural population)	1049	68.9262	1	2.795214	8.407935	30.99627	89.67651	99.5321	100	100
Electricity Access (% of Urban population)	1102	88.48147	25.98465	50.3	61.8	83.60088	98.45939	99.93	100	100

Additionally, there is still the possibility of endogeneity, by which we may have omitted variables which both have an effect on the dependent variable and are correlated with the independent variables (e.g. ‘ethnic fractionalisation’), despite our efforts to enact rigorous control variables. This is further complicated by the potential for ‘reverse causality’ - by which increased access to electricity increases the amount of aid to the recipient’s energy sector - and ‘simultaneity’ - by which the direction of

¹ Except for the KOF Globalisation index, which was sourced from the KOF website.

causality runs in both directions. In light of similar concerns, Gopalan and Rajan (2016) conduct dynamic estimations of their models using both Arellano-Bond Generalised Method of Moments (GMM), as in Arellano and Bover (1995) as well as system GMM as in Blundell and Bond (1998), as additional robustness checks. Future research should apply these robustness checks in order to verify results for this case.

To address these limitations, we employ several steps. The first is running all our regressions using country fixed effects specifications, which helps to mitigate against omitted variables, by allowing us to control for unobserved country specific effects. To strengthen this, we apply control variables which are widely used in foreign aid literature, and sourced the data from credible databases. Finally, we use a lagged indicator for aid disbursements, which reduces the potential for joint endogeneity.

4. Empirical Results

Our empirical investigation was split into three areas: the effect of ODA received for the energy sector on access to electricity for the whole population; and then, on access to electricity in rural and urban populations respectively. In all models, country fixed effects are accounted for, and robust standard errors are clustered.

Table 2 shows the country fixed effects estimates measuring the link between ODA disbursements and access to electricity for the overall population of a country. The coefficients for both $LnAid_t$ and $LnAid_{t-1}$ are statistically significant at the 5% and 10% level respectively. They are also both economically significant, with a 1% increase in ODA, and ODA in the previous year being associated with a 0.665 and 0.551 percentage point increase in electricity coverage, respectively. As for $LnAid_t/Population$, we find that it was economically significant, but not statistically significant, even at the 10% level. Considering the other two specifications' statistical significance, these results are surprising.

Dependent Variable: Access to Electricity (% of population)	(1) $LnAid_t$	(2) $LnAid_t/Pop$	(3) $LnAid_{t-1}$
$LnAid_t$	0.665** (0.297)		
$LnAid_t/Pop$		0.346 (0.278)	
$LnAid_{t-1}$			0.551* (0.312)
GNI per Capita	-0.000162 (0.000290)	-0.000168 (0.000291)	-0.000167 (0.000290)
$Lninflation$	0.395* (0.221)	0.411* (0.222)	0.436* (0.221)
Net FDI Inflows	0.00318 (0.0666)	0.00556 (0.0676)	0.00966 (0.0677)
Government Expenditure	-0.142 (0.125)	-0.134 (0.126)	-0.130 (0.125)
Population Density	0.0330 (0.0362)	0.0338 (0.0365)	0.0325 (0.0365)
Age Dependency Ratio	-1.115*** (0.218)	-1.119*** (0.219)	-1.121*** (0.218)
Government Effectiveness	-0.197 (1.565)	-0.225 (1.573)	-0.270 (1.566)
Regulatory Quality	1.717 (2.026)	1.712 (2.031)	1.828 (2.035)
Rule of Law	-0.948 (1.589)	-0.811 (1.586)	-0.773 (1.553)
KOF Globalization Index	-0.337* (0.180)	-0.341* (0.181)	-0.335* (0.179)
Constant	161.3** (19.54)	164.2** (19.51)	162.1** (19.61)
Number of Observations	710	710	712

Tables 3 and 4 show the country fixed effects estimates measuring the link between ODA disbursements and access to electricity for rural and urban populations respectively. These tables demonstrate potential heterogeneity in the effects of ODA on access to electricity within populations. Our regressions for rural populations (table 3) shows a similar result as in table 2 - the coefficients for both $LnAid_t$ and $LnAid_{t-1}$ are both economically and statistically significant (both at the 5% level). Interestingly, both these coefficients are noticeably higher than for the total population - a 1% increase in aid suggests a rise in electricity coverage of rural population of more than one percentage point, with the delayed effect being more pronounced. However, once again, though the coefficient on $LnAid_t/Population$ is economically significant, it is not statistically significant, even at the 10% level. Further, none of our regressions measuring the impact of ODA on urban population's electricity coverage are found to be statistically significant, and the coefficient of $LnAid_t/Population$ is negative, which suggests a negative relationship between aid disbursements and electricity coverage.

Table 3: Fixed Effects Estimates - Impact Of Energy Aid Disbursements On Access To Electricity for Rural Populations

Dependent Variable: Access to Electricity (% of population)			
	(1) $LnAid_t$	(2) $LnAid_t/Pop$	(3) $LnAid_{t-1}$
$LnAid_t$	1.095** (0.531)		
$LnAid_t/Pop$		0.731 (0.489)	
$LnAid_{t-1}$			1.562** (0.631)
GNI per Capita	-0.000516 (0.000623)	-0.000524 (0.000623)	-0.000487 (0.000622)
$Lninflation$	0.00629 (0.426)	0.0224 (0.428)	0.0912 (0.430)
Net FDI Inflows	0.0765 (0.114)	0.0800 (0.115)	0.0990 (0.113)
Government Expenditure	0.0214 (0.115)	0.0330 (0.115)	0.0291 (0.108)
KOF Globalization Index	-0.466 (0.294)	-0.476 (0.295)	-0.467 (0.289)
Population Density	0.0511 (0.0519)	0.0523 (0.0524)	0.0480 (0.0525)
Age Dependency Ratio	-0.281 (0.287)	-0.288 (0.289)	-0.295 (0.285)
Government Effectiveness	0.728 (2.713)	0.690 (2.725)	0.670 (2.667)
Regulatory Quality	6.331* (3.453)	6.327* (3.437)	6.683* (3.421)
Rule of Law	-2.267 (2.280)	-2.096 (2.268)	-2.081 (2.270)
Constant	105.2*** (21.73)	109.8*** (21.14)	103.7*** (21.84)
Number of Observations	710	710	712
Adjusted R - Squared	0.0837	0.0795	0.0932
Country Fixed Effects	YES	YES	YES

Note: Robust standard errors clustered for countries in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Fixed Effects Estimates - Impact Of Energy Aid Disbursements On Access To Electricity for Urban Populations

Dependent Variable: Access to Electricity (% of population)			
	(1) $LnAid_t$	(2) $LnAid_t/Pop$	(3) $LnAid_{t-1}$
$LnAid_t$	0.0486 (0.269)		
$LnAid_t/Pop$		-0.173 (0.264)	
$LnAid_{t-1}$			0.275 (0.226)
GNI per Capita	-0.000331 (0.000425)	-0.000328 (0.000426)	-0.00030 (0.00042)
$Lninflation$	0.0396 (0.182)	0.0490 (0.183)	0.0465 (0.187)
Net FDI Inflows	-0.0392 (0.0389)	-0.0375 (0.0391)	-0.0362 (0.0377)
Government Expenditure	-0.0721 (0.121)	-0.0655 (0.120)	-0.0737 (0.119)
KOF Globalization Index	-0.580*** (0.181)	-0.579*** (0.181)	-0.580*** (0.180)
Population Density	0.00792 (0.0151)	0.00810 (0.0152)	0.00728 (0.0153)
Age Dependency Ratio	-0.212 (0.152)	-0.210 (0.153)	-0.213 (0.152)
Government Effectiveness	0.832 (1.123)	0.803 (1.126)	0.883 (1.120)
Regulatory Quality	0.335 (1.738)	0.346 (1.732)	0.370 (1.738)
Rule of Law	-0.584 (1.743)	-0.480 (1.739)	-0.751 (1.739)
Constant	137.9*** (15.74)	138.5*** (15.63)	136.5*** (15.50)
Number of Observations	746	746	748
Adjusted R - Squared	0.0986	0.0992	0.0992
Country Fixed Effects	YES	YES	YES

Note: Robust standard errors clustered for countries in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Additionally, our regressions uncovered some interesting relationships between electricity coverage and our control variables, though these were not robust to all specifications. Of these, the most robust finding was a negative relationship between the KOF globalisation index and access to electricity, which suggests that as a country becomes more globalised, this is associated with a decrease in electricity coverage. This was statistically significant across all regressions for tables 2 and 3.

Thus, overall we find that there is a positive and statistically significant relationship between aid and electricity coverage, and that this is robust to multiple specifications. However, questions arise regarding the consistent lack of statistical significance associated with the specification measuring the impact of per capita aid. We discuss the potential explanations in the next section.

5. Discussion

The results analysed above suggest that: aid is associated with a positive impact on electricity access through contemporaneous and lagged channels; the effect of aid on electricity access through per capita channels is not statistically significant for any specification; the effect of aid through contemporaneous and lagged channels is statistically significant for electricity access of rural populations, but not urban ones; and there is a negative relationship between the KOF Globalisation Index and electricity coverage, robust to all specifications in tables 2 and 4. We discuss each of these findings in turn.

Regressions in table 2 suggest that aid is associated with a positive effect on access to electricity for a population through both contemporaneous ($LnAid_t$) and lagged ($LnAid_{t-1}$) channels. This suggests that ODA can have both long term and immediate effects in improving electricity access. More specifically, the significance of the $LnAid_{t-1}$ specification suggests that there may be a delayed impact in converting ODA received into higher electricity coverage for the entire population. It is especially applicable for the energy sector, where new infrastructure projects have long lead times, such as the construction of power plants and power grids. This is combined with the significance of the $LnAid_t$ variant of ODA, which suggests that there are also immediate effects. This may take the form of extra ODA easing liquidity constraints in ongoing projects (including those close to completion), aiding spending on maintenance or immediate sources, such as generators.

All our regressions also point to the statistical insignificance for all $LnAid_t/Population$ specifications – a somewhat surprising finding, given the significance of other ODA variations. One potential explanation may be that it is overall ODA received, rather than ODA received per capita, that matters in improving access to electricity. As high levels of investment and capital costs are required to set up new energy infrastructure, this suggests that reaching a high absolute threshold of ODA is more important than high ODA per head, for increasing electricity coverage. This is especially the case in projects involving renewable energy sources as means to increase electricity coverage, which typically incur high fixed costs (Donovan and Nunez 2012; Steffen and Waidelich 2022).

Tables 3 and 4 point to another interesting finding: that $LnAid_t$ and $LnAid_{t-1}$ were economically and statistically significant for rural, but not for urban populations. A possible explanation could be that there is a lower baseline of electricity coverage for rural populations. Table 1 illustrates this, where most notably, the differences in urban and rural populations' access to electricity peaks at over 50 percentage points at the 25th percentile. Thus, there is a higher marginal effect in rural areas as shown by the higher coefficients for $LnAid_t$ and $LnAid_{t-1}$ in table 3 compared to table 4. This

is consistent with findings in Gopalan and Rajan (2016), for the Water Supply and Sanitation sector.

Furthermore, within the rural population, the magnitude of the delayed effect was larger than the contemporaneous effect. Urmee et al. (2009) offers potential insight, finding that rural electrification can often be delayed as a result of poor day-to-day maintenance, lack of funding for consumers and after-sales services. In the context of our investigation's findings, this suggests that although the marginal effect of ODA to the energy sector is large in increasing access to electricity for the rural population, the manifestation of this effect can be delayed.

Our findings that higher trade openness characterised by the KOF Index, being associated with lower access to electricity, seems to contradict the literature on this issue, which broadly identifies that higher trade openness is associated with greater access to electricity (Fotio et al. 2023; Koengkan et al. 2019; Nounba and Ngueta 2023). However, the negative relationship we found may be the result of capital flight leading to lower investment in electrification (and thus lower access). Alternatively, redistribution of resources and investment towards other sectors that are better facilitated by the higher level of globalisation could have also contributed to the negative relationship found. Nonetheless, we do not rule out limitations in our data which may have led to this observed coefficient.

6. Conclusion

In conclusion, our paper empirically examines the effectiveness of foreign aid in enhancing access to electricity, a crucial aspect of the 7th SDG. Through the data procured from the OECD-DAC and the World Bank database, we employed the country fixed effects model to empirically estimate the effect of aid disbursement on access to electricity to the overall, rural, and urban population. Though the data is limited to after 2013 and the possibility of measurement errors, we took steps to mitigate the limitation from endogeneity and omitted variables. The results suggest that aid has a positive and statistically significant impact on access to electricity to the overall and rural population in instant and delayed conditions. On the other hand, it is not statistically significant to conclude that the urban population experiences the same impact on access to electricity from aid to the energy sector. Therefore, it is advised that aid donors target rural areas for energy aid to accomplish the 7th SDG. Our empirical study yielded conclusive results on the positive impact of aid on access to electricity; however, this only covers one aspect of the 7th SDG. As sustainability is a significant part of the SDGs, future studies must explore the effectiveness of foreign aid on the development of sustainable energy.

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